How NISAR Pro sees water change: a visual walkthrough

Bihar Ganga plain, NISAR track 98, frame 14, July to October 2026 · Aditya Narayan Nayak, Team RunLayer Labs, NISAR Pro Systems & SAR Analytics, Kundura, Odisha, India

> This document follows the data step by step: how one radar image becomes a water map, each pair of dates with real images, L-band and S-band together, and a time-lapse with the main observations. Every number is read from the result files in this package by `scripts/build_walkthrough_doc.py`. The figures come from `scripts/make_walkthrough.py`, using the same six NISAR files.

Part 1. The physics, the code and the 7-stage pipeline

The physics

- **Radar brings its own light.** NISAR sends microwave pulses and records the echo. Microwaves pass through cloud, so it works in the monsoon, day or night, when optical satellites only see cloud.
- **Calm water is a mirror.** The pulse bounces away from the satellite, so water looks dark. Rough ground, crops and buildings scatter energy back, so they look bright. The method looks for places that turned dark.
- **Two wavelengths.** L-band has a 24.2 cm wave (1.239 GHz) and S-band a 9.4 cm wave (3.2 GHz). A surface looks smooth when its bumps are much smaller than the wave, so the same sand bank or field can look smooth (dark) to L-band and rough (bright) to S-band.
- **HH and HV.** HH means the radar sends horizontal waves and listens for horizontal echoes. HV means it listens for vertical echoes; plants and rough surfaces turn part of the wave, so HV is stronger there. Water is detected with HH; HV is used in the fusion rules.
- **Decibels (dB).** Echo strength varies by thousands of times, so we use a log scale. Water is usually below −13 dB, land around −10 to 0 dB.

The code: who does what

One NISAR image layer is 1.16 billion numbers (4.63 GB). Opening it the normal way loads all of that into memory at once. The Rust engine never does that:

- `main.rs` reads the command (which two files, which band) and runs the steps in order.
- `windowed.rs` opens the HDF5 file and decodes only the compressed 512 × 512 chunks that are needed, never the whole layer.
- `nisar_parser.rs` reads the image in strips of 1,024 rows (33 strips) and averages every 8 × 8 block into one 80 m pixel, so only a 72 MB grid of 4,212 × 4,293 pixels stays in memory.
- `flood_detect.rs` runs stages 1 to 7 below.
- `io.rs` writes the results: GeoTIFF class maps and GeoJSON shapes with real map coordinates, plus a JSON report.

Each pair of dates takes 44 to 65 seconds with a peak of 1.87 to 2.32 GB of memory on a 4-core workstation. The L+S fusion, the Sentinel-2 checks and these figures are Python scripts in `scripts/`.

The 7 stages, one by one (L-band, 31 August → 12 September)

Before the stages: 10 m to 80 m. Each NISAR pixel is 10 m. While reading a strip, the engine takes every 8 × 8 block (64 pixels), keeps the values that are valid (power ≥ 0.0001), and averages them if at least 32 are valid: $\bar{P} = (1/n) \times \Sigma P_i$. It averages power, not dB, because power is the physical quantity; averaging dB would weight dark pixels too much. One 80 m pixel covers 0.0064 km² (1.58 acres).

Stage 1. Check the pair. The two dates must hold the same measurement: the same polarization term (HHHH) from the same product type (GCOV). Comparing HH on one date with HV on another would show a change that is not real. We checked this for every pair from the file names and the HDF5 layer paths.

Stage 2. Convert to decibels and mark no data. $\sigma^0 = 10 \times \log_{10}(\bar{P})$. Power below 0.0001 (−40 dB) is below what the radar can measure, so it is marked as no data. A pixel must be valid on both dates. Near the edge of the image the echo fades out and can look as dark as water, so the engine removes a strip 40 pixels (3.2 km) wide next to every no-data area. This is why the maps have a clean border and why a few small no-data spots inside the scene leave 6.5 km square gaps.

Stage 3. Remove speckle. Radar images are grainy: even a uniform field gives a mix of brighter and darker pixels. For every pixel the engine takes the 3 × 3 window around it (9 values), sorts the valid ones and keeps the middle value. A median is used, not a mean, because one very bright or very dark pixel cannot pull the middle value. It is done on both dates before comparing them.

Stage 4. Measure the change. $\Delta = \mathrm{dB(after)} - \mathrm{dB(before)}$, which is the same as $10 \times \log_{10}(\mathrm{P_after} / \mathrm{P_before})$. Because it is a ratio, anything that affects both dates in the same way, such as ground slope or a calibration offset, cancels out. Negative Δ means the place got darker; −3 dB means the echo halved, −10 dB means it dropped to a tenth.

Stage 5. Find water that was already there. A pixel darker than −13.0 dB on both dates (after the median) is stable water. These pixels are taken out of the next stages, because water that was already there is not new water.

Stage 6a. Choose the thresholds. The engine computes Otsu's threshold from the change histogram: it tries every cut point and picks the one that best separates the pixels into two groups, $\sigma^2_B = \omega_0 \times \omega_1 \times (\mu_0 - \mu_1)^2$, using only changes between −25 and 0 dB in 256 bins. Here it gave -3.26 dB for L-band and -3.47 dB for S-band. A safety limit of −3.0 dB stops a weak cut (for example on a windy day), and the seed must be at most −5.0 dB: $\mathrm{T_seed} = \min(-5.0, \min(\mathrm{Otsu}, -3.0)) = -5.0\ \mathrm{dB}$. So in these scenes Otsu is reported but does not change the map.

Stage 6b. Grow new water. Every pixel with Δ below −5.0 dB that is not stable water is a seed. From each seed the engine checks its 8 neighbours; a neighbour joins if its Δ is below −2.5 dB, and the search repeats until nothing more joins. One strict threshold would miss the edges of new water; one loose threshold would pick up noise everywhere. With two, a weak drop only counts if it touches a strong one.

Stage 6c. Clean up. First an opening: a pixel survives only if all 8 neighbours are also new water, then every survivor gets its 3 × 3 block back. This removes single noisy pixels and thin lines, but also any new water narrower than 3 pixels (about 240 m). Then any connected group smaller than 8 pixels (0.0512 km², 12.65 acres) is removed.

Stage 7. Give each pixel one class. 1 = stable water (stage 5). 2 = new water with Δ below the seed threshold (high confidence). 3 = the rest of the grown new water (medium confidence). 5 = water loss: below −13.0 dB before, at or above −13.0 dB after, and Δ of at least +3 dB. 0 = everything else. Area is just counting: $\mathrm{area} = N \times 0.0064\ \mathrm{km^2}$, and acres = km² / 0.004046856.

Five real pixels through the stages

Each row is a real 80 m pixel from the window in Figure 1, the pixel of that class nearest the window centre.

Pixel (lat, lon)	Power 31 Aug → 12 Sep	Stage 2 dB	Stage 3 median dB	Stage 4 Δ	Engine result
New water, high 25.3276°N, 86.7683°E	0.0923 → 0.0394	-10.35 → -14.05	-10.35 → -15.46	-5.12 dB	class 2
New water, medium 25.329°N, 86.7699°E	0.1069 → 0.0369	-9.71 → -14.33	-8.61 → -12.81	-4.20 dB	class 3
Stable water 25.3485°N, 86.7841°E	0.0369 → 0.0194	-14.33 → -17.13	-13.76 → -15.67	-1.92 dB	class 1
Water loss 25.389°N, 86.7674°E	0.0177 → 0.025	-17.51 → -16.01	-13.44 → -10.02	+3.41 dB	class 5
Land 25.35°N, 86.777°E	0.2772 → 0.1958	-5.57 → -7.08	-6.09 → -6.54	-0.46 dB	class 0

- The **high-confidence** pixel: the raw values drop by -3.70 dB, but after the median the drop is -5.12 dB. The neighbourhood is darker than the single pixel, which makes it a seed.
- The **medium-confidence** pixel dropped -4.2 dB: not enough to start a region, but enough to join one next to it.
- The **water-loss** pixel is still −16.0 dB as a single raw value on 12 September, but its 3 × 3 median is -10.02 dB, so the neighbourhood has become land. Stage 3 changed the answer here.

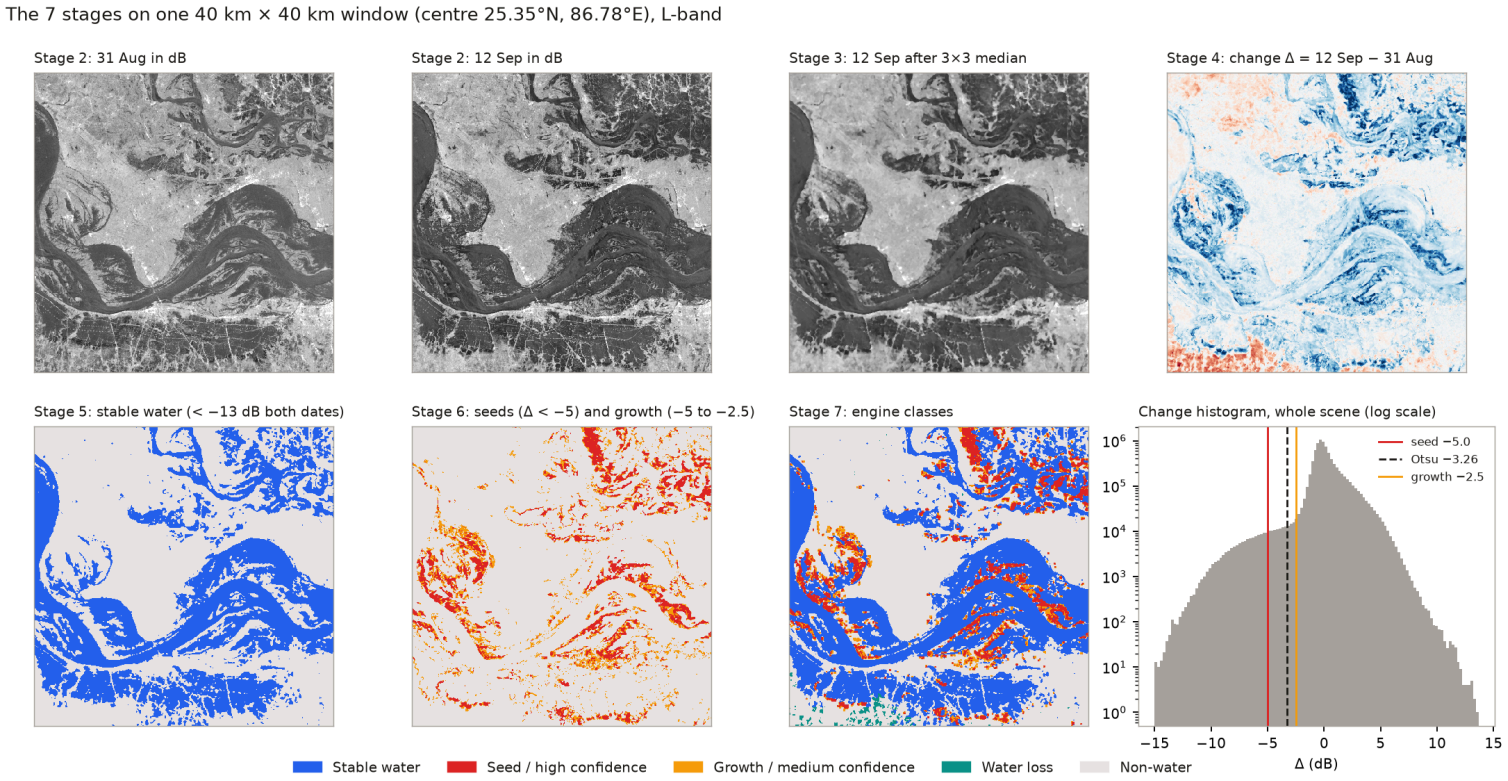


Figure 1. The stages on one 40 km × 40 km window centred at 25.35°N, 86.777°E, chosen automatically as the window with the most change. The images in stages 2 to 6 are recomputed for this figure; stage 7 is the engine's real output. Check: all 16,289 engine high-confidence pixels in this window also have Δ below −5 dB in our recomputation (100%).

How to read the histogram. Most pixels sit near Δ = 0 (nothing changed). The long left tail is the places that got much darker. The seed line (−5 dB) cuts far into that tail, so a pixel needs a strong drop to start a new-water region.

Part 2. L-band, 26 July → 31 August

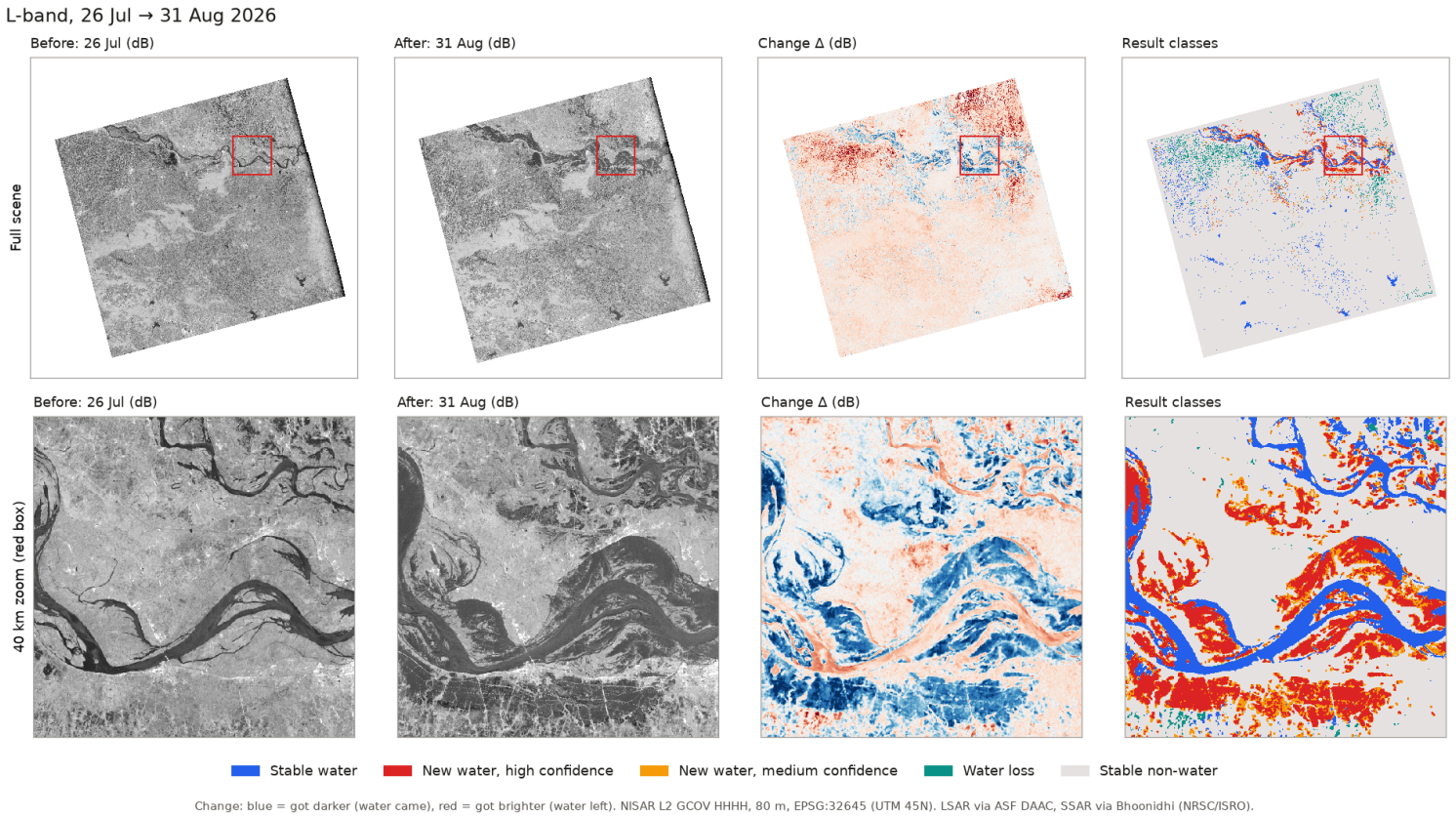


Figure 2. Top: full scene. Bottom: the 40 km window (red box). Columns: before, after, change, result.

- Water before 3,689.59 km² (911,718 acres), after 3,914.16 km² (967,210 acres).
- New water 1,695.81 km² (419,044 acres), of which 922.73 km² (228,012 acres) high confidence. Water loss 1,471.24 km² (363,551 acres).
- **What we see:** by the river (bottom row) the dark area spreads on both banks, and 425.29 km² (105,091 acres) of new water appears in this window alone. Out in the western farmland, many places that were dark on 26 July are brighter on 31 August, which shows up as water loss.
- **What it means:** the water did not just grow; it moved. It gathered along the river while scattered dark patches in the fields went away.

Part 3. L-band, 31 August → 12 September

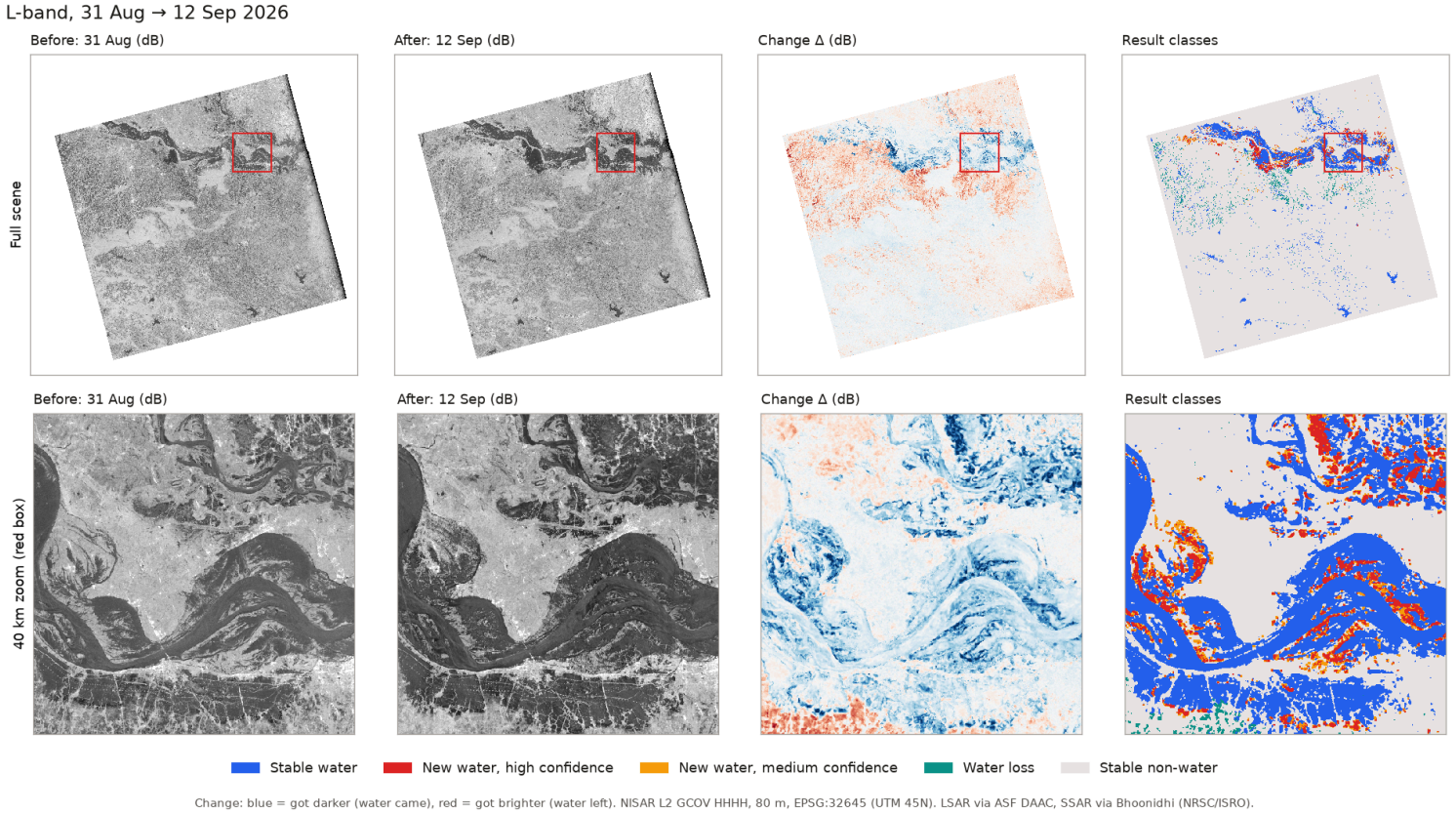


Figure 3. Same layout as Figure 2.

- Water before 4,245.03 km² (1,048,970 acres), after 4,521.28 km² (1,117,233 acres).
- New water 1,166.21 km² (288,177 acres) (713.74 km² (176,369 acres) high confidence). Water loss 889.96 km² (219,914 acres).
- **What we see:** near the river the blue (darker) areas keep spreading into side channels and low land, adding 162.76 km² (40,219 acres) in the window. Large farmland areas to the west turn brighter (red).
- The right edge of the full-scene images has a dark strip. That is the edge of the radar swath, not water, and it is why stage 2 cuts 40 pixels next to every no-data edge.

Part 4. L-band, 12 September → 18 October (recession)

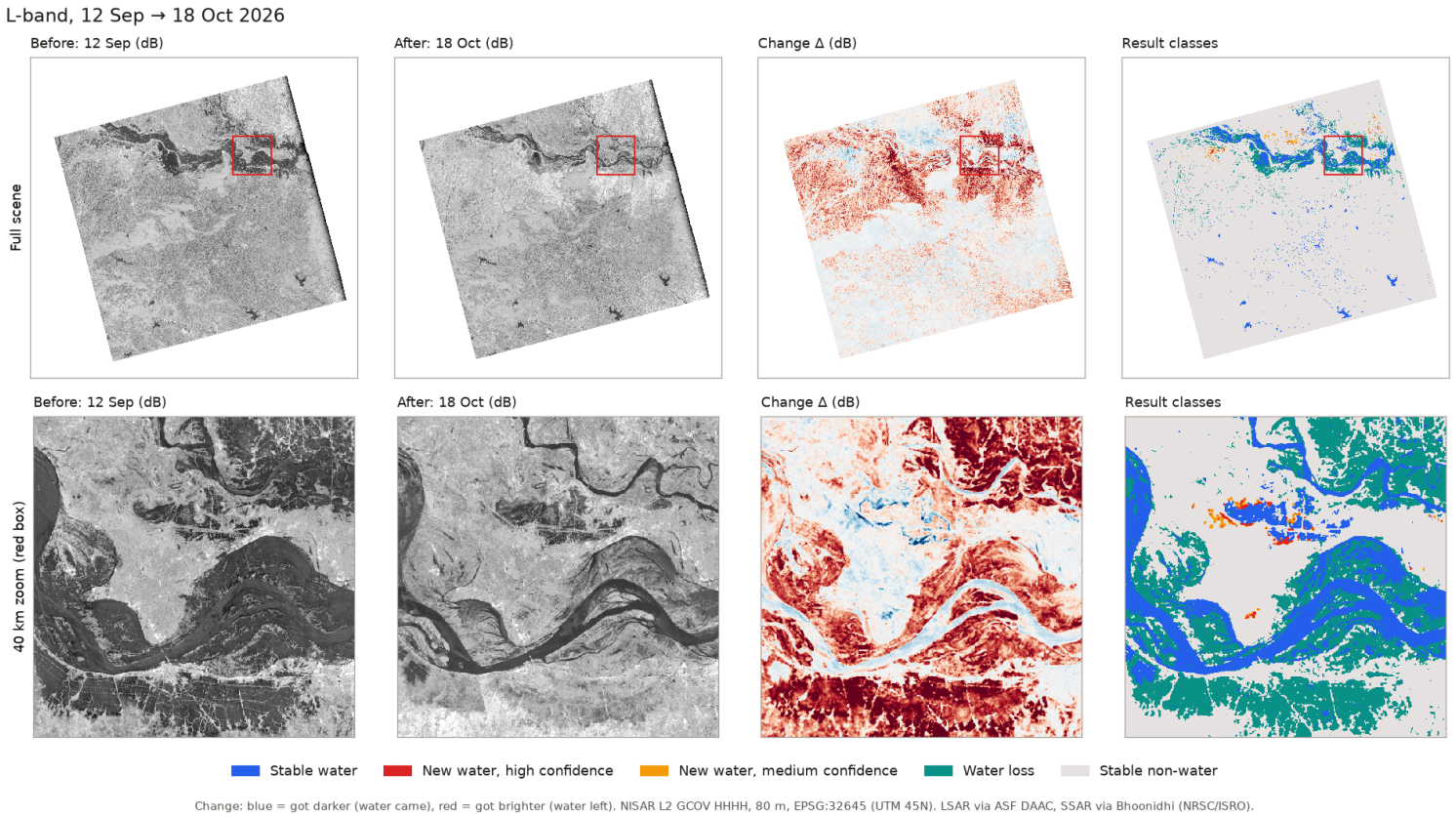


Figure 4. Same layout as Figure 2.

- Water before 4,598.52 km² (1,136,319 acres), after 2,286.19 km² (564,930 acres).
- Water loss 2,551.94 km² (630,598 acres), ten times the new water of 239.60 km² (59,206 acres).
- **What we see:** the change map is mostly red: places got brighter as water left. In the window, 514.50 km² (127,136 acres) turned from water back to land, and the river shrinks back towards its main channels.

Part 5. S-band, 31 August → 12 September

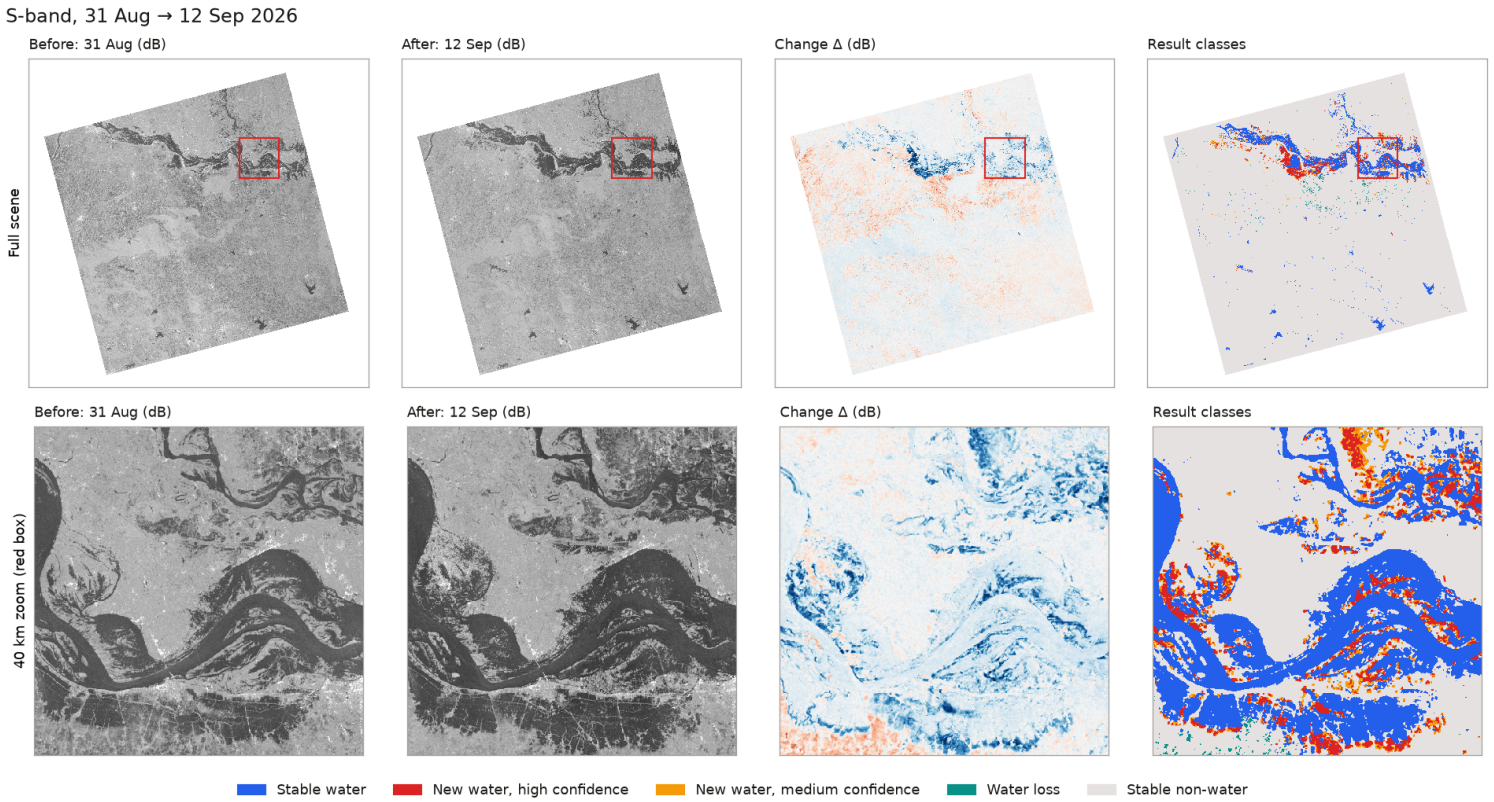


Figure 5. Same pipeline and layout, S-band. Only the thresholds differ (stable water −12.5 dB, seed −4.8 dB, growth −2.2 dB), because the shorter wave sees surfaces as rougher.

- Water before 2,794.56 km² (690,551 acres), after 3,752.28 km² (927,209 acres).
- New water 1,174.89 km² (290,322 acres), almost the same as L-band's 1,166.21 km² (288,177 acres).
- But stable water is 2,577.39 km² (636,887 acres) against L-band's 3,355.07 km² (829,056 acres), and water loss 217.17 km² (53,664 acres) against 889.96 km² (219,914 acres).
- **What it means:** both bands agree on where new water came, but L-band calls much more area "water" to begin with. Part 6 checks who is right.

These numbers use each band's own footprint. The fair three-way comparison in Part 6 uses the area both bands cover.

Part 6. Putting L and S together (fusion)

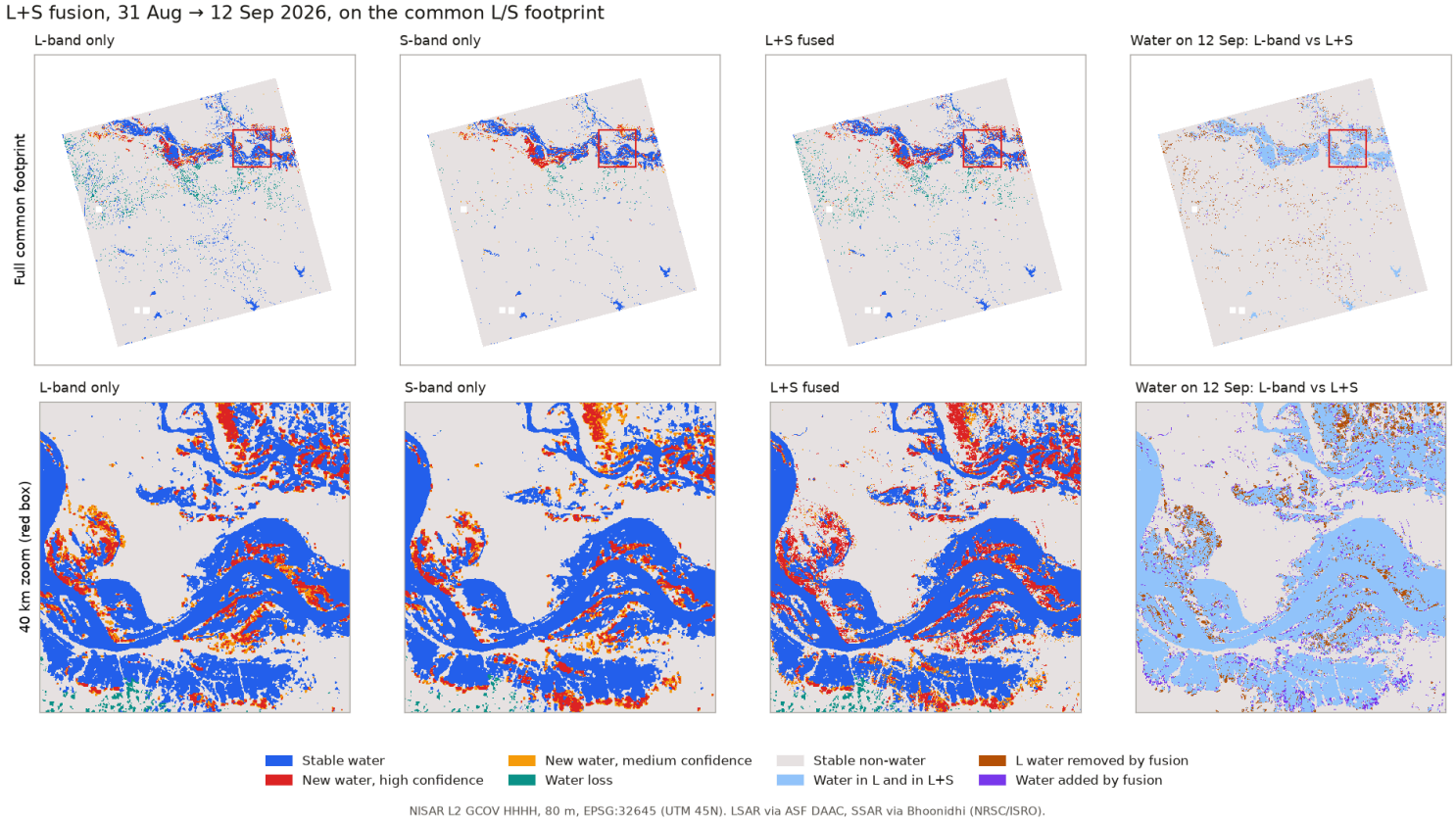


Figure 6. Classes for L-band, S-band and L+S on the common footprint, and how the 12 September water map changed between L-band and L+S.

Both grids line up exactly on 80 m pixels, so each pixel goes through four if/then rules:

1. **Sandbar rule:** dark in L (HH < −13 dB) but bright in S (HH ≥ −10.5 dB) means not stable water.
2. **Wind rule:** very dark L-band HH and HV on both dates counts as stable water even if S-band missed it.
3. **Canopy rule:** a strong L-band drop with low HV is a new-water candidate (not verified; optical data cannot see under plants).
4. **S-band gate:** every new-water candidate must also be dark in S-band on 12 September (HH < −10.5 dB).

Common footprint, 31 Aug → 12 Sep	Stable water	New water	Water loss
L-band	3,314.83 km² (819,112 acres)	1,137.14 km² (280,993 acres)	888.84 km² (219,637 acres)
S-band	2,546.37 km² (629,222 acres)	1,174.89 km² (290,322 acres)	214.98 km² (53,123 acres)
L+S	2,652.29 km² (655,395 acres)	1,337.15 km² (330,417 acres)	723.37 km² (178,749 acres)

- Compared with L-band, fusion removed 847.43 km² (209,405 acres) of L-band's 12 September water and added 362.24 km² (89,511 acres); 3,542.31 km² (875,324 acres) stayed water in both.
- Switching rules off one at a time: without the S-band gate, new water would be 1,675.85 km² (414,112 acres); without the canopy rule, 1,113.61 km² (275,179 acres).

Who is right? We checked against Sentinel-2, an optical satellite that passed about 5 hours after NISAR on 12 September, using only cloud-free pixels (a pixel is water if NDWI ≥ 0).

F1 score (1.0 = perfect agreement)	L-band	S-band	L+S
Water, 12 Sep	0.859	0.891	0.891
Water, 31 Aug	0.807	0.874	0.834
New water, 31 Aug → 12 Sep	0.555	0.652	0.637

The pixels that were dark in L-band but bright in S-band were water only 8.7% of the time on 12 September, against 80.3% where both bands were dark. So S-band was right to remove them. They were not all sand, though: on 12 September most of them were vegetation.

Part 7. Time-lapse: how the water changed over the season

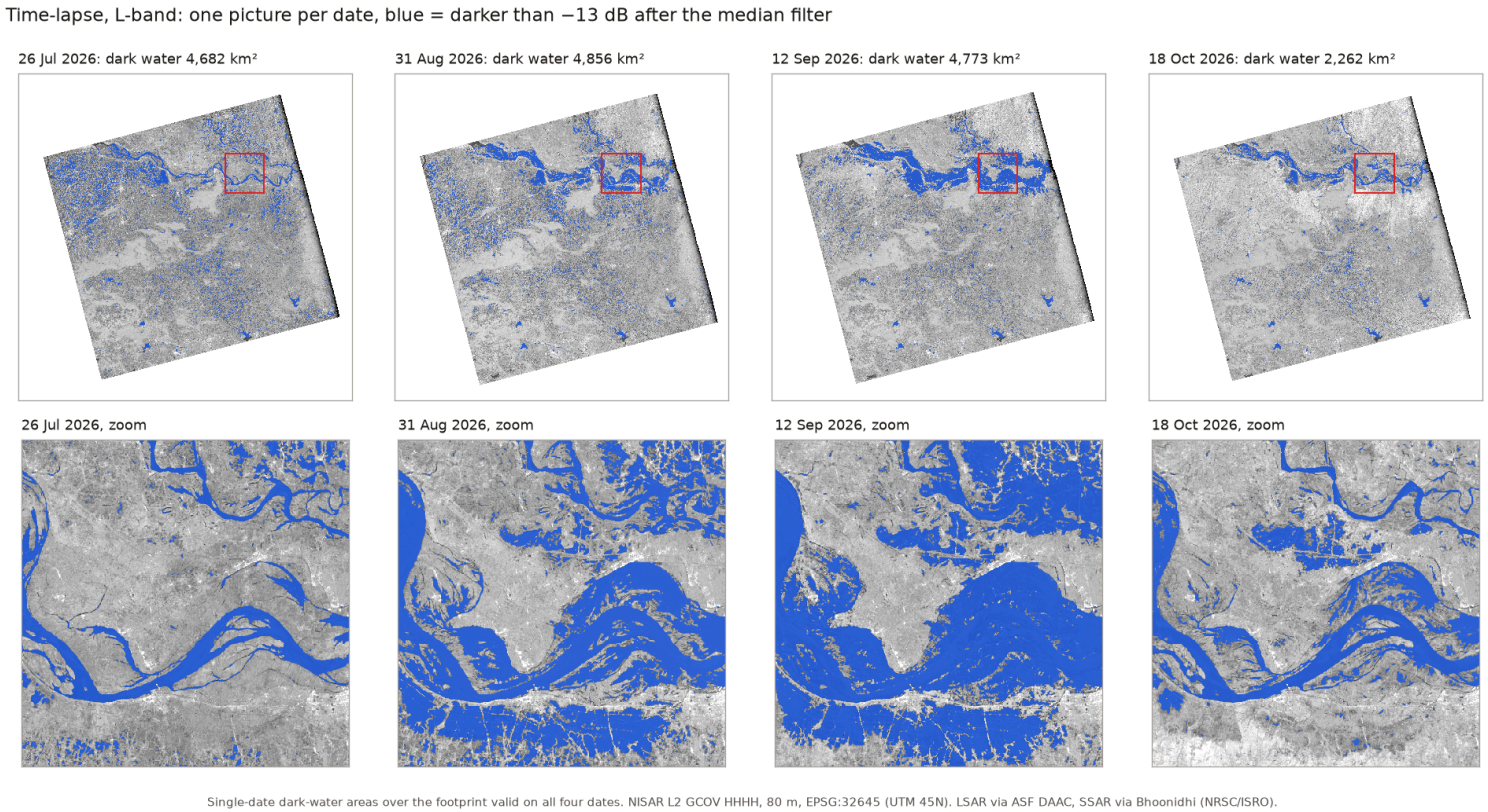
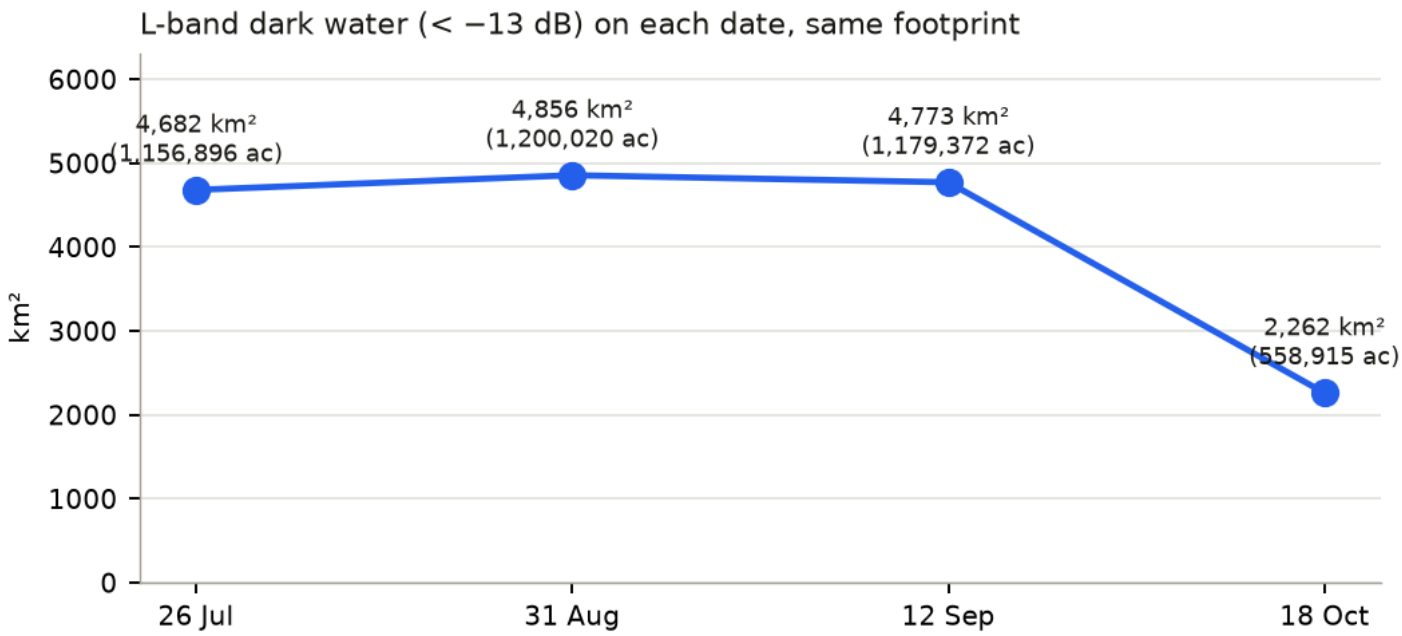


Figure 7. One L-band picture per date. Blue: darker than −13 dB after the median filter. This is a simple single-date view, not the change-detection result. An animated version is `explainer/timelapse_L_band.gif`.



Dark water (< −13 dB)	26 Jul	31 Aug	12 Sep	18 Oct
40 km window by the river	213 km²	617 km²	812 km²	299 km²
Rest of the scene	4,469 km²	4,240 km²	3,961 km²	1,963 km²
Whole footprint (55,807.39 km² (13,790,308 acres))	4,682 km²	4,856 km²	4,773 km²	2,262 km²

Acres: whole footprint 26 Jul 1,156,896, 31 Aug 1,200,020, 12 Sep 1,179,372, 18 Oct 558,915.

Part 8. Main observations

1. **Near the river, water grew a lot and then went back.** In the 40 km window, dark water went from 212.90 km² (52,609 acres) on 26 July to 811.83 km² (200,608 acres) on 12 September, about 3.8 times as much, and fell to 298.95 km² (73,872 acres) by 18 October.
2. **Across the whole scene the total hardly changed until October.** It stayed between 4,682 and 4,856 km² from July to September, because scattered dark patches in the farmland faded while the river area grew. Then it dropped by 53% by 18 October.
3. **Water moves, it does not only grow.** Every pair has both new water and water loss. Between 31 August and 12 September, L-band found 1,166.21 km² (288,177 acres) of new water and 889.96 km² (219,914 acres) of loss.
4. **New water away from the usual water.** 1,282.78 km² (316,982 acres) of L+S new water lies outside any water seen on 26 July or 31 August. The problem statement calls this potential temporary surface inundation.
5. **L-band sees more "water" than there is.** Against Sentinel-2, L-band had the most false alarms. S-band was the most accurate single band.
6. **Fusion fixes most of L-band's mistakes** but did not beat S-band alone in our tests.

> **What we cannot say.** Radar shows where water appeared or left, not why. We make no claim that any change was a flood, rain or irrigation. Optical checks only cover cloud-free pixels (about 14% of the scene on 12 September) and cannot see water under crops. Thresholds were set by hand. "Dark water" in Part 7 can include other smooth, dark surfaces.

Words used

Backscatter	The echo strength the radar receives from a place.
GCOV	The NISAR product we used: map-ready echo strength for each polarization.
Pixel	One 80 m × 80 m square, 0.0064 km² (1.58 acres).
Footprint	The area the satellite actually imaged and we analysed.
F1 score	Agreement between two water maps, from 0 to 1. It counts both missed water and false water.
Sentinel-2, NDWI	A European optical satellite, and a green-vs-infrared index that is high over water.